

EMVCo 3-D Secure 2.0 & Payment Network Tokenisation Specification

Frictionless Authentication Flows, Risk-Based Telemetry & DPAN Token Lifecycle Architecture

Standard Setter:	EMVCo LLC (Visa, Mastercard, Amex, Discover, JCB, UnionPay)
Specification ID:	EMVCO-3DS-2.3
Classification:	Cardholder Authentication & Tokenisation Architecture
Verification:	Integrated in Ariya Sarrafzadeh Production Architecture

1. 3DS 2.0 Frictionless vs. Challenge Flows

The protocol shares rich merchant and device contextual telemetry (100+ data elements) with the card issuer's Access Control Server (ACS), enabling risk engines to authenticate over 90% of transactions seamlessly without SMS OTP friction.

2. Biometric Liability Shift Mechanics

Upon successful cryptographic authentication verification (CAVV / AAV cryptogram), legal liability for fraudulent chargebacks shifts definitively from the merchant to the card issuer.

3. Network Tokenisation (DPAN vs FPAN)

Replaces 16-digit Funding Primary Account Numbers (FPAN) with Device/Merchant Primary Account Numbers (DPAN). Each transaction generates a dynamic one-time cryptogram (DTVV), rendering compromised database records useless.

4. Token Lifecycle Automation

Automated webhook integrations with Token Requestors track card lifecycle events (card expiry updates, lost/stolen replacements) with zero merchant checkout disruption.

Systems Verification Notice: This authoritative specification establishes the invariant compliance baseline for production fintech switches, settlement reconciliation engines, and double-entry ledger implementations documented across the portfolio.